

A. Syllabus

Course No	Title of the Course	Course Structure	Pre-Requisite
FCPH..	Physics 2: Quantum Physics	3L-1T-0P	None
COURSE CONTENT			
Unit I			
Origin of Quantum Physics: Blackbody radiation, Rayleigh scattering, Inadequacy of classical physics, Need for quantum physics, Historical overview, Plank's hypothesis, Quantization of energy, Origin of quantum theory, Photoelectric effect, Compton scattering.			
Unit II			
Wave-Particle Duality: de Broglie hypothesis – matter waves, Wave-particle duality, Wave-packets, Phase velocity and Group velocity, Experimental evidence of de Broglie's hypothesis: Davisson–Germer experiment, Matter waves for macroscopic objects, Complementarity principle, Heisenberg uncertainty principle.			
Unit III			
Quantum Physics and Schrodinger Wave Equation: Born's interpretation of the wave function, Probability density and Normalization, Expectation values, Free-particle wave function, Momentum and energy operators, Eigen functions and Eigen values, Schrodinger equation – time-dependent and time-independent.			
Unit IV			
Applications of Schrodinger Equation: Solution of stationary-state schrodinger equation for one dimensional problems – Particle in a box, Square-well potential and potential step, Quantum tunnelling effect and related examples like Alpha-decay, Field-ionization and Scanning Tunnelling Microscope.			
Unit V			
Quantum Mechanics in Solids : Free electron theory of metals, Fermi level, Density of states, Energy and momentum diagrams, Effective mass of electron, Bloch's theorem for particles in a periodic potential, Kronig-Penney model and origin of energy bands, Classification of solids			
Text Books:			
1. Concepts of modern physics by Arthur Beiser (McGraw Hill). 2. Quantum Mechanics by A. Ghatak and S. Lokanathan (Macmillan India Ltd). 3. Quantum Mechanics: Concept and Applications by Nouredine Zettili (Wiley). 4. Introduction to quantum mechanics by D. J. Griffiths (Pearson Education, IInd Edition).			